

Sunny Joshi
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[Website](#) | [GitHub](#)
Nationality – British
Indian OCI holder

Professional Summary

Machine Learning Engineer with 6+ years of experience in deep learning, computer vision, NLP, and MLOps. Strong background in AI research, model deployment, and high-performance computing. Expertise in Transformers, CNNs, contrastive learning, and generative models. Proven track record of building scalable ML solutions in fintech, healthcare, and geospatial domains.

Technical Skills

Programming & Tools: Python, PyTorch, TensorFlow, Bash, Docker, PyTorch-Lightning, Weights & Biases, Slurm, HPC, Poetry, scikit-learn, XGBoost, Hugging Face, DeepSpeed, Kubernetes (basic)
Machine Learning & AI: Transformers, CNNs, RNN/LSTMs, MAE, VAE, Language Models, Contrastive Learning, Diffusion Models
Cloud & MLOps: AWS (SageMaker, EC2, RDS), Azure, CI/CD, Model Versioning, Feature Engineering, Data Pipelines

Industry & Research Experience

Senior Machine Learning Engineer | Digehealth (March 2024 – Present)

- Working on multimodal (speech and vision) models improving gut health in post-operative patients.

PhD Researcher (Earth Observation & ML) | The University of Edinburgh, UK (Oct 2023 – Dec 2024, discontinued)

- Developed multimodal contextual dataset combining Landsat and Sentinel data, migrated to [Amazon Open Data Sponsorship](#).
- Developed distributed training pipelines and docker files to train on HPC and Kubernetes clusters.
- Designed self-supervised MAE-based regression models for domain generalization in Earth Observation.
- Investigated large-scale foundation models for geospatial vision tasks.

Applied Science Intern (Internship as a requirement of the CDT) | Roku, Cambridge, UK (Jul 2024 – Sep 2024)

- Generated synthetic datasets using Stable Diffusion, ControlNet, inpainting/outpainting to enhance YOLOX model training and developed pipeline to train on AWS EC2.
- Led test/train dataset distribution strategy, improving model generalization.

Machine Learning Engineer | Cambrian Robotics (Jan 2024 – Feb 2024)

- Developed Ray Tuning Pipeline for architecture design and HP-Tuning
- Prototyped architectures for improving segmentation and object detection performance in object picking software

Senior AIML Platform Engineer | GSK, London, UK (May 2023 - Oct 2023)

- Designed scalable ML pipelines for bio-model deployment on Azure HPC & cloud services.
- Developed an end-to-end DNA sequence classification pipeline, showcasing ML platform capabilities, and investigated fine-tuning, pruning and compression techniques.

Senior Machine Learning Engineer | Informa Markets, London, UK (Sep 2022 – Mar 2023)

- Developed LDA-guided BERT for document embeddings, improving NLP model accuracy by 25%.
- Built MLOps infrastructure (data versioning, model versioning, dependency management, CI/CD).
- Engineered dynamic pricing forecasting models using advanced data pipelines.

Senior Associate I, Data Science I AIG, London, UK (Apr 2021 – Aug 2022)

- Developed and maintained time-series forecasting models deployed on AWS.
- Built geospatial Django pipelines, migrating legacy systems to DVC-based versioning.

Machine Learning Engineer I AIG, London, UK (Jun 2019 – Mar 2021)

- Created feature engineering and modelling tools for DeepAR-based time-series forecasting.
- Implemented AWS RDS-based reproducibility pipelines, streamlining ML workflows.

Graduate Software Developer (NLP Research) I Sage Group Plc, Newcastle, UK (Feb 2018 – Mar 2019)

- Enhanced chatbot performance using semi-supervised memory LSTMs, CRFs.
- Benchmarked in-house NLP models against RASA & Alterra frameworks.

Key Projects & Contributions

- **Open Data Contribution:** Created an open-source, multi-sensor satellite dataset, now part of AWS Open Data Registry.
- **Deep Learning for Vision:** Implemented MAE-based contrastive and self-supervised learning models for domain adaptation in geospatial imagery.
- **Synthetic Data Generation: Applied Diffusion models** to generate high-quality synthetic datasets for AI training.
- **MLOps Engineering:** Designed **automated ML pipelines**, ensuring reproducibility and scalability in production.

Education

- Master of Science in AI, The University of Edinburgh, UK (2016 - 2017)
- Master of Technology in Software Engineering, Manipal Institute of Technology, India (2012 - 2014)
- Bachelor of Technology in Computer Engineering, Mody Institute of Technology and Science, India (2008 - 2012)

Links & Profiles

- GitHub: github.com/sunny1401
 - Website: <https://sunny1401.github.io>
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